

PROJECT FINANCE MODEL GUIDE

40 MW Hyperscale-Ready Data Centre

Behind-the-Meter 90 MWp Solar PV + 60 MWh BESS | Greater Accra, Ghana

A Plain-English Guide to Project Finance Concepts, Model Structure and Every Key Formula

Written for a financial / accounting professional who is new to project finance

Companion document to: Ghana_DataCentre_Solar_BESS_Project_Finance_Model.xlsx

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1. What Is Project Finance?

1.1 The Core Idea

Project finance is a way of funding a single, large, capital-intensive asset — a toll road, a power plant, a data centre — by lending against the cash flows that the asset itself will generate, rather than against the general creditworthiness of the company that builds it. In corporate finance, a bank lends to a company and looks at the whole company's balance sheet, its other businesses, and its overall credit rating. In project finance, the bank lends to a single-purpose entity that owns only this one project, and the loan is repaid only from what that project earns.

This matters enormously for how a model like this one is built. Because there is no diversified parent company standing behind the debt, lenders cannot rely on "the company will figure it out if this project has a bad year." Every risk has to be identified, quantified, and either allocated to a party who can bear it, or sized into the debt structure so the project can absorb it and still pay lenders on time. That is the entire purpose of a project finance model: to prove, number by number, that the project's own cash flow is sufficient to service its debt through every foreseeable scenario.

1.2 Why Use Project Finance Instead of Ordinary Corporate Borrowing?

- Scale: projects like this one cost hundreds of millions of dollars — often more than a single sponsor wants to, or can, put on its own balance sheet.
- Risk containment ("non-recourse" or "limited recourse"): if the project fails, lenders' claims are generally limited to the project's own assets and cash flows, not the sponsor's other businesses. This ring-fencing is the defining feature of project finance.
- Risk-sharing: construction risk, operating risk, market risk and financing risk are each allocated by contract to whichever party is best placed to manage them (see 1.4 below).
- Higher leverage than a typical corporate loan: because the debt is secured directly against a well-contracted, predictable cash flow stream, lenders will often accept gearing (debt as a % of total cost) that would be considered aggressive for an ordinary corporate loan — this model targets 65%, though as you'll see in Section 7, the actual achievable gearing here is lower.

1.3 The Special Purpose Vehicle (SPV)

Almost every project financing is done through a Special Purpose Vehicle — a newly created legal entity whose only asset is this one project. The SPV signs the construction contract, the operating contract, the offtake (customer) contracts, and the loan agreement. Sponsors (the equity investors) own the SPV; lenders lend to the SPV; and the whole financial model in the accompanying workbook is, in effect, modelling the SPV's own income statement, balance sheet, and cash flow statement from financial close through to the end of the debt tenor.

1.4 Who's In The Room: The Key Parties

Party	Role
Sponsor(s) / Equity	Develop the project, contribute equity capital, and receive distributions once debt service and reserve requirements are met. They bear the residual risk and reward.
Senior Lenders	Provide the largest tranche of debt, senior in priority of repayment. In this model, a single sculpted senior facility.
SPV	The ring-fenced legal entity that owns the project, signs every contract, and is the direct counterparty to lenders.

EPC Contractor	Engineering, Procurement & Construction contractor — typically engaged under a fixed-price, fixed-date contract so construction cost and schedule risk sits with the contractor, not the SPV or lenders.
O&M Contractor	Operates and maintains the asset post-COD, usually under a long-term contract with performance guarantees.
Offtaker(s) / Customers	The counterparties who pay for the project's output — here, the hyperscale/enterprise data centre customers under 15-year take-or-pay contracts. Their creditworthiness is central to lenders' comfort.
Independent Engineer / Technical Advisor	Reviews the technical design, construction progress and generation/yield assumptions on lenders' behalf.
Model Auditor	An independent firm that checks the financial model for mechanical accuracy and logic before financial close — this is exactly the kind of review the Checks-Audit tab in this workbook is designed to pass.

1.5 The Project Lifecycle

- **Development:** feasibility studies, permitting, contracting (EPC, O&M, offtake), and financial structuring — before any construction capital is spent.
- **Financial Close:** the point at which all financing and project contracts are signed and conditions precedent are satisfied; debt and equity become available to draw.
- **Construction:** the EPC contractor builds the asset; debt and equity are drawn down to fund capex; interest accrues on drawn debt but is not yet paid in cash — it is capitalised (see Interest During Construction, Section 2).
- **Commercial Operation Date (COD):** the project is complete and begins generating revenue; this is when scheduled debt repayments typically begin. COD for this project is targeted for 1 July 2028.
- **Operations:** the project runs for the length of its offtake contracts and/or debt tenor (15 years of debt amortisation in this model), generating the cash flow that services debt and, after that, distributes to equity.
- **Maturity / Exit:** debt is fully repaid; sponsors may continue to operate the asset, sell it, or refinance.

2. Core Project Finance Concepts & Terminology

This section explains, in plain language, every core concept a corporate/accounting finance professional needs before opening the model. Each concept links directly to a tab or formula you'll meet in Sections 4 and 5.

2.1 CFADS — Cash Flow Available for Debt Service

CFADS is the single most important number in any project finance model. It is the cash the project generates in a period, before any financing costs, that is available to pay lenders. Unlike ordinary corporate EBITDA, CFADS is a cash concept — it deducts cash tax but is calculated before interest and principal repayments (those come next, in the waterfall).

$$\text{CFADS} = \text{EBITDA} - \text{Cash Tax}$$

(Working capital movements and maintenance capex would also be deducted in a fuller model; this project has none of material size outside of the reserve mechanisms modelled separately.)

2.2 DSCR — Debt Service Coverage Ratio

DSCR measures how many times over the project's cash flow could pay its debt service in a given period. It is the ratio lenders watch most closely, both as a sizing tool (how much debt can the cash flow support?) and as an ongoing covenant (if DSCR falls below a threshold, dividends to equity can be "locked up" i.e. suspended).

$$\text{DSCR} = \text{CFADS} \div \text{Total Debt Service (Interest + Principal) for the period}$$

This model targets a minimum DSCR of 1.35x — meaning CFADS must be at least 1.35 times debt service in every period. A DSCR of 1.0x would mean the project has exactly enough cash to pay debt service with nothing left over (and no margin for error); anything below 1.0x means the project cannot pay lenders in full from its own operating cash flow that period. Lenders build in a cushion (1.35x here) precisely so that a bad-but-plausible period doesn't tip the project into shortfall.

Lenders typically look at both the Minimum DSCR (the worst single period across the whole tenor — the binding constraint) and the Average DSCR (the overall health of the cover). Both appear on the Outputs & Dashboard and Checks-Audit tabs of this model.

2.3 LLCR and PLCR — Loan Life and Project Life Coverage Ratios

Where DSCR looks at a single period, LLCR and PLCR look forward across the remaining life of the loan (LLCR) or the remaining life of the project/contracts (PLCR). Both compare the present value of future CFADS to the current outstanding debt balance:

$$\begin{aligned} \text{LLCR} &= \text{PV(Remaining CFADS to loan maturity, discounted at the cost of debt)} \div \\ &\text{Outstanding Debt Balance} \\ \text{PLCR} &= \text{PV(Remaining CFADS to end of project life)} \div \text{Outstanding Debt Balance} \end{aligned}$$

A ratio above 1.0x means there is, in present-value terms, more than enough future cash flow to repay the outstanding debt. These ratios are common in lender term sheets as additional covenants alongside DSCR, though this model's primary sizing and covenant mechanic is DSCR-based sculpting (Section 2.6).

2.4 Gearing (Leverage)

Gearing is the proportion of the project's total cost funded by debt rather than equity:

$$\text{Gearing} = \text{Senior Debt} \div (\text{Senior Debt} + \text{Equity}) \quad [\text{or, as used in this model: Debt} \div \text{Hard Capex}]$$

Higher gearing means equity investors put in less capital and, if the project performs, earn a higher return on that smaller base (financial leverage). But it also means less cushion if things go wrong, and it requires the cash flow to be strong and reliable enough to service more debt at the target DSCR. This is why gearing is never simply chosen — it emerges from the tension between a target (65% in this brief) and what the cash flow can actually support at the required DSCR. Section 7 explains why, in this specific project, that tension resolves at roughly 36% rather than 65%.

2.5 IDC — Interest During Construction

During construction, the project is spending money (capex) but earning no revenue yet. Debt is still drawn to fund construction, though, and interest starts accruing on it immediately. Since there's no operating cash flow yet to pay that interest in cash, it is capitalised — added to the total project cost — rather than paid. This is Interest During Construction, and it is why the Total Project Cost is always larger than the sum of hard construction costs alone.

2.6 Debt Sculpting

An ordinary mortgage or corporate term loan is usually repaid on a flat annuity schedule — the same total payment every period. Project finance debt is different: because project cash flow is rarely flat (it often ramps up as a facility reaches full utilisation, or varies seasonally), lenders and sponsors "sculpt" the principal repayment schedule so that debt service in every period tracks a constant multiple of CFADS — the target DSCR — rather than being a fixed amount.

In practice this means: in periods where CFADS is strong, the project repays more principal; in periods where CFADS is weaker, it repays less. The result is that DSCR is held at exactly the target (1.35x here) in every single period — no period is more or less comfortable than any other — and the total amount of debt the project can support is maximised for a given target DSCR. This is precisely the mechanic built into the Debt & Financing tab of this model, and it's worth sitting with because it's the part of project finance modelling that differs most from ordinary corporate debt schedules.

2.7 DSRA — Debt Service Reserve Account

A DSRA is a cash reserve, funded upfront and topped up or released over time, that exists purely as a buffer: if CFADS in some future period is temporarily insufficient to cover debt service, the DSRA can be drawn to make up the shortfall, protecting the project from a technical default. Lenders commonly require a DSRA sized at some number of months of forward debt service — six months in this project's case — and it is funded before COD as part of the project's Sources & Uses.

2.8 The Cash Flow Waterfall

The waterfall (sometimes called the cash flow cascade) is the strict order in which the project's cash is applied each period. Nothing further down the waterfall can be paid until everything above it has been paid in full. A standard project finance waterfall — and the one used in this model — runs:

- 1. Revenue, less Operating Costs = EBITDA
- 2. Less Cash Tax = CFADS
- 3. Less Senior Debt Service (interest + scheduled principal)
- 4. Less DSRA funding (or plus DSRA release)
- 5. Less reserve account contributions (here, the BESS augmentation reserve) and any reserve shortfalls
- 6. = Cash Available for Distribution → paid to equity as distributions

Equity is deliberately last in the queue. That's the essence of the risk/return trade lenders and sponsors strike: lenders get paid first but earn a fixed return (interest); equity is paid last but keeps all the upside once every senior obligation is satisfied.

2.9 Reserve Accounts Beyond the DSRA

Many projects with equipment that has a shorter useful life than the debt tenor — batteries are the classic example — set aside cash over time into a dedicated reserve to fund a known future replacement or refurbishment cost, rather than being caught short when that cost arrives. This model includes a BESS Augmentation Reserve for exactly this reason: the battery system is expected to need a mid-life augmentation around Year 8, and the model funds a reserve for it from period 1 onward.

2.10 Circularity in Project Finance Models

A subtle but very common modelling problem: interest expense is usually tax-deductible, so it reduces taxable income, which increases after-tax cash flow (CFADS), which increases how much debt the project can support, which changes the interest expense — which is exactly the number you started with. This loop is called circularity, and if built literally into a spreadsheet it can send Excel into an endless recalculation loop unless "iterative calculation" is deliberately turned on.

This model uses a Circularity Breaker (an on/off switch on the Assumptions tab) to avoid that loop entirely: with the breaker ON (the default), cash tax is calculated without deducting interest, so CFADS — and therefore debt sizing — no longer depends on the debt itself. This keeps the whole model non-circular and reliably calculable with a single pass, at the cost of a small simplification (ignoring the interest tax shield). Section 5.6 shows exactly where this is implemented, and Section 7 explains how to switch it off if you want the fully circular version.

2.11 Base Case vs. Sensitivities and Downside Cases

A "Base Case" is the model's central, most-likely set of assumptions. But no lender finances a project on the Base Case alone — they stress-test it. This model's brief specifically calls for a live Best Case / Base Case / Worst Case switcher (Section 4.2), plus one-way sensitivities on individual variables like interest rates and capex (Section 4.10). The purpose is always the same: to confirm the project can still service its debt (DSCR stays at or above target) even when individual assumptions move against it.

2.12 Equity Returns: IRR, NPV and Multiple

Metric	What it means
Equity IRR	The discount rate at which the present value of all equity cash outflows (capital invested) equals the present value of all equity cash inflows (distributions received). It is the annualised, compounding rate of return earned on equity capital, accounting for the exact timing of every cash flow.
Equity NPV	The net present value of all equity cash flows, discounted at the sponsor's required rate of return (12% in this model). A positive NPV at that discount rate means the investment clears the sponsor's hurdle rate; negative means it doesn't.
Equity Multiple	Total cash distributed to equity, divided by total equity invested — a simple multiple-of-money return that ignores timing (unlike IRR).

3. This Project At A Glance

Before diving into the model itself, here is the specific project this workbook finances, and why its two halves — the data centre and the renewable power system — have to be modelled as one integrated whole rather than two separate businesses.

3.1 The Two Assets, One Project

The SPV owns two physically and financially linked assets:

- A 40 MW IT-capacity data centre (design PUE 1.25), leased to customers under 15-year capacity and energy contracts — 85% on a take-or-pay basis, 15% on shorter-term/spot terms.
- A dedicated, behind-the-meter 90 MWp single-axis-tracking solar array plus a 60 MWh (4-hour) battery energy storage system, built specifically to power the data centre directly rather than to sell power to the grid.

Because the renewable system sits "behind the meter" — i.e. wired directly to the data centre rather than exporting to the public grid — it earns no separate revenue of its own. Its entire economic contribution is to reduce the data centre's energy costs (compared to buying 100% of its power from the grid) and to support the project's sustainability credentials. This is exactly why the Operations tab (Section 4.5) models both assets in a single integrated flow, netting renewable generation against data-centre demand every period, rather than as two unrelated businesses.

3.2 Base Case Parameters

Parameter	Base Case Value
IT Capacity	40 MW
Design PUE	1.25
Utilisation ramp	55% (Yr 1) → 75% (Yr 2) → 90% (Yr 3+)
Capacity charge	\$18.50 / kW-month
Take-or-pay contracted share	85% (residual 15% on spot terms)
Solar capacity	90 MWp, single-axis tracking
Specific yield (P50)	1,680 kWh/kWp/yr
BESS	60 MWh (4-hr equivalent), 90% round-trip efficiency
Construction period	24 months (Jul-2026 → Jun-2028)
Target COD	1 July 2028
Debt tenor	15 years from COD
All-in interest rate	7.75%
Target minimum DSCR	1.35x
Target gearing	65%

4. Model Architecture — A Guided Tour of Every Tab

The workbook is built as thirteen linked tabs, each with a single, clear job. Formulas flow strictly downstream — the Assumptions tab feeds everything; Construction and Operations feed Debt & Financing; those feed the Waterfall; the Waterfall feeds Outputs. Nothing calculates "backwards" into an earlier tab (this is what keeps the whole model free of unwanted circularity — see Section 2.10).

1. Cover

The control panel. Contains the live Scenario selector (a dropdown: Base Case / Best Case / Worst Case) and the headline KPI tiles (Equity IRR, NPV, Multiple, DSCR, Gearing, Total Project Cost, COD) which pull live from the Outputs tab. This is the one cell — Cover!B7 — that drives the entire model when you change scenario.

2. Assumptions

Every input the model uses, organised by category (Data Centre, Renewable System, Costs, Financing, Macro & Tax, Timing). Each input has three editable columns — Base Case, Best Case, Worst Case — in blue font, plus a fourth "Active" column that automatically picks whichever case matches the Cover sheet's scenario selector. Every other tab in the model reads only from the Active column, never from Base/Best/Worst directly.

3. Timing & Calendar

The master date axis. Two calendars: a 24-month Construction Calendar (Month 1 to Month 24) and a 30-period Operations Calendar (Period 1 to Period 30, each period six months, spanning the 15-year debt tenor). Also computes, for each operating period, the escalation factor (for inflation) and the solar degradation factor — both of which every revenue and cost line downstream depends on.

4. Construction

Models the 24-month capex drawdown: how much is spent each month (via a smooth S-curve), how that spend is split between debt and equity, how interest during construction (IDC) accrues on the debt drawn, and a Sources & Uses reconciliation confirming the money needed exactly equals the money raised at COD.

5. Operations

The heart of the project's ongoing economics. Models data centre utilisation and energy demand, solar generation and BESS losses, the resulting renewable self-sufficiency percentage and grid-shortfall cost, and all data centre revenue and operating cost lines, rolling up to EBITDA — for all 30 operating periods.

6. Debt & Financing

Converts EBITDA into CFADS (via the tax calculation), sizes the senior debt facility (the smaller of a DSCR-sculpted capacity and the 65% gearing cap), and builds the full sculpted amortisation schedule — opening balance, interest, scheduled principal, closing balance and DSCR — for every period, plus the DSRA sizing.

7. Cash Flow Waterfall

Lays out the full priority-of-payments cascade described in Section 2.8, from revenue down to equity distributions, in one continuous view — this is the tab a lender's credit committee would read first.

8. Reserves

Detailed mechanics of the DSRA (opening/target/funding-or-release/closing balance each period) and the BESS Augmentation Reserve (periodic contributions, the one-off augmentation capex event, and any funding shortfall).

9. Outputs & Dashboard

Builds the full equity cash flow — 24 monthly construction outflows followed by 30 semi-annual operating distributions — and calculates Equity IRR, NPV, Multiple, and the headline credit metrics (Min/Avg DSCR, Actual Gearing, Total Project Cost) that feed the Cover sheet tiles.

10. Sensitivities & Scenarios

A live comparison table showing the currently active scenario's key metrics, plus indicative one-way sensitivities of Equity IRR to interest-rate and capex shocks, with guidance on how to run precise (non-approximated) sensitivities directly.

11. Checks-Audit

The independent quality-control tab: automated pass/fail checks for Sources=Uses, full debt repayment, minimum DSCR, no negative cash, timing consistency, and the live scenario switcher — plus a written narrative confirming the waterfall priority, dual-asset logic, and overall structure are correct, exactly as a model auditor's sign-off would read.

12. Appendix-Glossary

A quick-reference glossary split into Technical/Modelling terms and Non-Technical/Commercial terms — a condensed companion to Section 8 of this guide.

13. Charts

Visual summaries: the revenue/EBITDA build across all periods, the DSCR profile against its 1.35x target, the renewable-vs-grid energy supply mix, and the full equity cash flow profile.

5. Formula Deep-Dive — How Every Number Is Calculated

This section walks through every major formula block in the model, in the order cash actually flows through the project: assumptions → construction → operations → debt → waterfall → returns. Sheet and row references match the delivered workbook exactly, so you can follow along cell-by-cell.

5.1 The Live Scenario Switcher

Every row on the Assumptions tab has this formula in its "Active" column:

```
=INDEX(C7:E7, MATCH($C$3, {"Base Case", "Best Case", "Worst Case"}, 0))
```

MATCH finds the position (1, 2 or 3) of the current scenario name (pulled from Cover!B7 into Assumptions!C3) within the three-item list; INDEX then returns the corresponding value from that row's Base/Best/Worst columns. Because every single formula elsewhere in the model references only this Active column — never the Base/Best/Worst columns directly — changing one dropdown cell on the Cover sheet instantly recalculates the entire project: generation, revenue, debt sculpting, ratios, and returns, with zero duplicated logic anywhere.

5.2 Construction: The Capex S-Curve

Rather than a hand-typed, hardcoded spend profile, construction spend follows a formula-driven parabola that is slow at the start, peaks in the middle of the 24-month programme, and tapers at the end — a realistic mobilisation-to-completion shape:

```
Raw Weight(t) = t × (25 - t)                                for month t = 1 ... 24
Normalised Weight(t) = Raw Weight(t) ÷ ΣRaw Weight(1..24)
Monthly Capex Draw(t) = Total Hard Capex × Normalised Weight(t)
```

Because the weights are normalised to sum to exactly 100%, total drawn capex always equals total hard capex by construction, with no manual balancing needed.

5.3 Construction: Debt Drawdown and IDC

Debt and equity are drawn each month in proportion to the ratio between the final sculpted debt size (see Section 5.7) and total hard capex — not a flat 65% assumption. This is deliberate: the amount of debt the project can actually support is determined by its cash flow (via DSCR sculpting), and construction-phase drawdown has to reconcile with that same number, or the Sources & Uses would not balance.

```
Debt Draw(t) = Total Capex Draw(t) × (Final Sculpted Debt Size ÷ Total Hard Capex)
IDC(t) = Opening Cumulative Debt Balance(t) × (Annual Interest Rate ÷ 12)
Equity Draw(t) = Total Capex Draw(t) - Debt Draw(t) + IDC(t)
```

IDC is calculated monthly on the opening cumulative debt balance (the balance before that month's new draw), consistent with how interest genuinely accrues on debt already outstanding. IDC itself is funded by equity, not by further debt draws, which keeps the calculation straightforward and non-circular.

5.4 Operations: Data Centre Demand

```
Utilisation(t) = Year-1 / Year-2 / Year-3+ rate, selected by the operating year
IT Load (MW) = IT Capacity × Utilisation(t)
Facility Load (MW) = IT Load × PUE
Facility Energy Demand (MWh) = Facility Load × 182.625 days × 24 hours
```

182.625 days is exactly half of a 365.25-day year — the modelling convention used throughout for a semi-annual period. PUE (Power Usage Effectiveness) grosses up the pure IT load to total facility power draw — see the Glossary for the concept.

5.5 Operations: Renewable Generation, BESS Losses, and Grid Shortfall

```
Gross Solar Generation (MWh) = Solar Capacity (MWp) × Specific Yield (kWh/kWp/yr)
                                × (1 - Curtailment %) × Degradation Factor(t) ×
                                (182.625 ÷ 365.25)

BESS Round-Trip Loss (MWh) = Gross Generation × BESS Cycled Share × (1 - BESS Round-
Trip Efficiency)

Net Renewable Supply (MWh) = Gross Generation - BESS Round-Trip Loss

Grid Shortfall (MWh) = MAX(0, Facility Energy Demand - Net Renewable Supply)

Self-Sufficiency % = MIN(100%, Net Renewable Supply ÷ Facility Energy Demand)
```

Note the units: multiplying Solar Capacity in MWp directly by Specific Yield in kWh/kWp/yr already gives MWh/yr (the "k" and the "M" cancel through the 1,000x capacity scaling) — a common point of unit-conversion error in solar models, and one this model's build process specifically caught and corrected.

The BESS Cycled Share (30%, an explicit modelling assumption on the Assumptions tab) represents the portion of total generation that is charged into and discharged from the battery rather than consumed directly and instantaneously; only that cycled portion incurs the round-trip efficiency loss.

Because a 60 MWh / 4-hour battery is small relative to a continuous 24/7, 40 MW data centre load, self-sufficiency in this Base Case comes out at roughly 35–59% across the operating periods — a genuine, realistic finding for this system size rather than a modelling flaw (Section 7 discusses this in depth).

5.6 Operations: Revenue and Cost Build

```
Capacity Revenue (Take-or-Pay) = IT Capacity (kW) × 85% × Capacity Charge ($/kW-
month)
                                × Escalation(t) × Months in Period

Capacity Revenue (Spot) = IT Capacity (kW) × 15% × Utilisation(t) × Capacity Charge
                                × Escalation(t) × Months in Period

Energy Revenue = Facility Energy Demand (MWh) × 1,000 × Energy Charge ($/kWh) ×
Escalation(t)

Grid Backup Cost = Grid Shortfall (MWh) × 1,000 × Grid Backup Price ($/kWh) ×
Escalation(t)
```

The 85% take-or-pay portion of capacity revenue is earned regardless of utilisation — that is precisely what "take-or-pay" means contractually (Glossary, Section 8). The 15% residual/spot portion only earns revenue in proportion to actual utilisation, reflecting its shorter-term, less contractually certain nature.

A modelling note worth flagging explicitly: the brief specified that the energy charge should be set at "an effective rate consistent with the renewable supply cost plus margin," but did not give an exact figure. This model sets Energy Charge and Grid Backup Price as explicit, editable Assumptions-tab inputs, calibrated so the energy charge sits comfortably above the blended cost of supply (a mix of near-zero-marginal-cost renewable generation and grid backup purchases) — but these should be replaced with real quoted/contracted figures as soon as they're available.

5.7 Debt & Financing: Tax, CFADS and the Circularity Breaker

```

Tax Depreciation – Data Centre(t) = IF(t ≤ 2 × Dep. Life, DC Capex ÷ (2 × Dep. Life),
0)
Tax Depreciation – Renewable(t) = IF(t ≤ 2 × Dep. Life, RE Capex ÷ (2 × Dep. Life),
0)

Taxable Income(t) = MAX(0, EBITDA(t) – Tax Depreciation DC(t) – Tax Depreciation
RE(t))
Cash Tax(t) = Taxable Income(t) × Corporate Tax Rate

CFADS(t) = EBITDA(t) – Cash Tax(t)

```

The "× 2" converts an annual straight-line depreciation life (11 years for the data centre, 12 for renewables, per the brief) into a semi-annual period count, since the model operates on 30 semi-annual periods. With the Circularity Breaker engaged (the default), interest expense is deliberately excluded from the tax calculation — this is what makes CFADS entirely independent of the debt structure, and is the single design choice that keeps this whole model free of circular references (Section 2.10).

5.8 Debt & Financing: Sizing the Senior Debt Facility

Because CFADS does not depend on debt at all (per 5.7), the maximum debt the project's cash flow can support at the target DSCR can be calculated directly, in one pass, with no iteration required:

```

DSCR-Sculpted Debt Capacity = Σ [ (CFADS(t) ÷ Target DSCR) ÷ (1 + Semi-Annual
Interest Rate)^t ] for t = 1..30

Gearing Cap = Target Gearing % × Total Hard Capex

Final Debt Size = MIN(DSCR-Sculpted Debt Capacity, Gearing Cap)

```

This is exactly the sculpting logic from Section 2.6, expressed as a single present-value calculation: it finds the debt amount that, if serviced at exactly the target DSCR every period, would be precisely paid off using the discount rate equal to the cost of debt. The model then takes whichever is smaller — this DSCR-based capacity, or the sponsor's 65% gearing target — as the actual amount financed, and records which constraint is binding (see the "Binding Constraint" cell on the Debt & Financing tab).

5.9 Debt & Financing: The Sculpted Amortisation Schedule

```

Opening Balance(t) = Closing Balance(t-1) [Opening Balance(1) = Final
Debt Size]
Interest(t) = Opening Balance(t) × Semi-Annual Interest Rate
Target Debt Service(t) = CFADS(t) ÷ Target DSCR
Scheduled Principal(t) = MIN(Opening Balance(t), MAX(0, Target Debt Service(t) –
Interest(t)))
[final period (t=30): Principal = Opening Balance(t), a full true-up to zero]
Total Debt Service(t) = Interest(t) + Scheduled Principal(t)
Closing Balance(t) = Opening Balance(t) – Scheduled Principal(t)
DSCR(t) = CFADS(t) ÷ Total Debt Service(t)

```

Because the Final Debt Size in 5.8 was derived as exactly the present value of the target debt service stream, this schedule self-amortises to zero by the final period without any manual plugging — the small "true-up" in the final period simply mops up any residual rounding. Every intermediate period's DSCR comes out at exactly the 1.35x target, which you can verify directly on the Debt & Financing tab.

5.10 Reserves: DSRA Sizing

```

DSRA Target(t) = Total Debt Service(t+1) × (DSRA Months ÷ 6) [0 in the final
period]

```

```

DSRA Funding / (Release)(t) = DSRA Target(t) - DSRA Opening Balance(t)
DSRA Closing Balance(t) = DSRA Opening Balance(t) + Funding/(Release)(t)

```

The DSRA is always sized to hold enough cash to cover the next period's debt service (scaled by however many months of cover the sponsor's financing terms require — six months, i.e. one full period, in this case). It is funded upfront at COD from the Sources & Uses, and thereafter tops up or releases automatically as the target moves with the amortisation schedule.

5.11 The Waterfall, Formalised

```

EBITDA = Data Centre Revenue - Operating Costs
CFADS = EBITDA - Cash Tax
Cash Flow after Debt Service = CFADS - Total Debt Service
Cash Available for Distribution = MAX(0, Cash Flow after Debt Service
                                   - DSRA Funding/(Release)
                                   - Augmentation Reserve Contribution
                                   - Augmentation Capex Shortfall)
Equity Distribution = Cash Available for Distribution

```

The MAX(0, ...) guard ensures the model never displays a negative distribution — if the waterfall genuinely could not fund a downstream item, the Checks-Audit tab's "no negative cash" check would flag it rather than let a negative number silently appear.

5.12 Equity Returns: XIRR and XNPV

Because construction cash flows are monthly and operating cash flows are semi-annual — irregular, unevenly spaced intervals — the model uses Excel's date-aware XIRR and XNPV functions rather than the simpler IRR/NPV functions, which assume evenly spaced periods:

```

Equity IRR = XIRR(Equity Cash Flow Range, Corresponding Date Range)
Equity NPV = XNPV(Equity Discount Rate, Equity Cash Flow Range, Corresponding Date Range)
Equity Multiple = Total Positive Distributions ÷ Total Equity Invested (absolute value)

```

The Equity Cash Flow Range itself is built by concatenating 24 monthly construction-phase equity outflows (negative, from the Construction tab) with 30 semi-annual operating-phase equity distributions (positive, from the Waterfall tab) — a single continuous 54-column timeline on the Outputs & Dashboard tab, each column tagged with its real calendar date.

5.13 Sensitivities: The Approximation Method

The Sensitivities & Scenarios tab includes two indicative one-way sensitivity tables (Equity IRR vs. an interest-rate shock, and vs. a capex shock) built as simple, clearly-labelled elasticity approximations around the current Active Scenario's IRR, rather than re-running the full model for each shock size:

```

IRR(rate shock) ≈ Base IRR - (Actual Gearing × 0.9 × Rate Shock)
IRR(capex shock) ≈ Base IRR - (1.6 × Capex Shock %)

```

These are deliberately flagged on the tab itself as approximations, not exact recalculations — a genuine one-way Excel Data Table, or simply overwriting the relevant Assumptions cell directly and reading the live Outputs, will always be more precise. The formulas exist to give a fast, directional feel for sensitivity without materially increasing the model's calculation load.

6. How To Use The Model

6.1 Switching Scenarios

- Go to the Cover tab, click cell B7, and choose Base Case, Best Case, or Worst Case from the dropdown.
- Every KPI tile on the Cover tab, every ratio on Checks-Audit, and every number on Outputs & Dashboard will recalculate instantly — there is nothing else to change or re-run.
- To compare all three cases side by side, switch the dropdown to each case in turn and record the Sensitivities & Scenarios tab's "Current Active Scenario" column each time (or set up an Excel Data Table keyed to Cover!B7 for a fully automated three-way comparison).

6.2 Editing Assumptions

- All editable inputs live on the Assumptions tab in blue font on a pale green background — this is the only place you should type new numbers.
- Every other cell in the model is a formula. If you need to test something the Assumptions tab doesn't cover (e.g. a different BESS cycled-share assumption), it is safest to add a new row on the Assumptions tab and reference it, rather than hardcoding a number into a calculation tab.
- Changing a Base Case cell changes the Base Case only; Best/Worst Case cells are independent and won't move unless you edit them directly.

6.3 Reading the Outputs

- Start on the Cover tab for the headline picture, then go to Outputs & Dashboard for the full equity cash flow and the DSCR/gearing detail behind the headlines.
- The Debt & Financing tab shows the full period-by-period sculpted amortisation and DSCR — this is the tab to check if a DSCR-related output looks unexpected.
- The Cash Flow Waterfall tab shows exactly how cash moves from revenue down to equity distributions each period, in the same order a lender's term sheet would specify.

6.4 Running Your Own Precise Sensitivity

For anything more precise than the approximated tables on the Sensitivities tab (Section 5.13), simply overwrite the relevant Base Case cell on the Assumptions tab directly — e.g. change the interest rate from 7.75% to 8.50% — and read the updated Equity IRR, NPV, and DSCR straight off the Outputs & Dashboard tab. Because the whole model is a single forward-flowing calculation with no circularity (Section 2.10), there is no need to force a recalculation or worry about stale values — the answer updates the moment you enter the number.

6.5 What To Check If You're Auditing This Model

- Open the Checks-Audit tab first — every mechanical check (Sources=Uses, full debt repayment, minimum DSCR, no negative cash, timing consistency, live scenario switcher) is automated and will read OK or ERROR.
- Read the "Notes on Model Design & Key Findings" block at the bottom of that same tab — it discloses every simplification and judgement call made in building the model (the circularity breaker, the invented energy/grid-backup pricing, the S-curve shape, the BESS cycling assumption, and the base-case economics finding), so nothing is a hidden surprise.
- Trace any single number back to its source: every formula in this workbook is a direct cell reference back through Operations → Assumptions, with no pasted values anywhere in the calculation chain.

7. Key Findings, Assumptions & Limitations

7.1 The Headline Finding: Base Case Equity IRR Is Negative

Under the Base Case as specified, Equity IRR comes out at roughly -11.5% , with an Equity Multiple of about $0.27\times$ — meaning equity investors would, on these assumptions, get back only around a quarter of their invested capital over the life of the project. This is a genuine output of the model given the stated assumptions, not a calculation error, and it is driven by two compounding factors:

- Debt capacity, not the 65% gearing target, is the binding constraint. At a $1.35\times$ minimum DSCR, the project's own cash flow can only support roughly 36% gearing — well short of the 65% sponsors were targeting. That pushes a much larger share of the $\sim\$508\text{m}$ Total Project Cost onto equity than the brief's target capital structure implied.
- The renewable system's self-sufficiency is modest (roughly $35\text{--}59\%$ across the operating periods) because a $60\text{ MWh} / 4\text{-hour}$ battery cannot firm a 90 MWp solar array against a continuous $24/7, 40\text{ MW}$ data centre load. The majority of the facility's energy therefore has to be backed up at a cost that leaves only a thin margin over the modelled energy charge.

Neither of these is a flaw in the model's mechanics — both are legitimate economic findings that a real project sponsor would want to know before financial close, and both are exactly the kind of insight a bankable model is supposed to surface.

7.2 Levers Most Likely to Restore Bankability

- A higher energy and/or capacity charge (i.e. testing whether the offtake contracts can support pricing closer to what covers the true blended cost of supply plus an adequate margin).
- A larger BESS (more hours of storage) to lift renewable self-sufficiency and reduce reliance on grid backup.
- Additional firm capacity — a grid power purchase agreement or a gas peaker — specifically to reduce the grid-backup cost drag rather than relying on solar+small-BESS alone.
- Lower capex (the Sensitivities tab shows that roughly a 10% capex reduction alone is enough to flip Equity IRR positive under the current pricing assumptions).

7.3 Modelling Simplifications — Fully Disclosed

Simplification	Detail
Circularity breaker (default ON)	Cash tax excludes interest deductibility, so CFADS and debt sizing are independent of the debt itself. This keeps the model reliably calculable without enabling Excel's iterative calculation. To run the fully circular, interest-tax-shielded version: set Assumptions!"Circularity Breaker" to 0, enable File > Options > Formulas > Enable Iterative Calculation, and update the tax formula on Debt & Financing to deduct interest.
Energy Charge & Grid Backup Price	Not specified in the original brief; set by the modeller at $\$0.155/\text{kWh}$ and $\$0.14\text{--}0.16/\text{kWh}$ respectively, consistent with the brief's instruction to price energy at "renewable supply cost plus margin." Both are ordinary Assumptions-tab inputs and should be replaced with real contracted/quoted figures.
Semi-annual period length	182.625 days (exactly half of a 365.25 -day year) used consistently for every period-based calculation.

Capex S-curve shape	A formula-driven parabola ($t \times (25-t)$), normalised to 100% rather than a hand-typed monthly profile — realistic in shape, but not tied to an actual EPC contractor's drawdown schedule.
BESS cycled-generation share	30% of gross solar generation assumed to cycle through the battery (and therefore incur round-trip losses) — an explicit, editable modelling assumption rather than a value derived from a detailed dispatch simulation.
BESS augmentation event	Modelled as a single lump-sum event at operating period 16 (~Year 8), sized at 20% of renewable capex — a simplification of what would likely be a more gradual or multi-stage augmentation programme in reality.
Debt drawdown timing during construction	Debt and equity are drawn pro-rata to total monthly spend at a fixed ratio (the final sculpted debt size $\div$ hard capex), rather than a more complex "debt-first" or "equity-first" drawdown sequencing sometimes seen in real facility agreements.

7.4 Confirmed Correct on Audit

The Checks-Audit tab explicitly confirms — and this guide re-confirms — that the following core mechanics are structurally correct and consistent with standard project finance practice:

- Waterfall priority: Revenue → Operating Costs → EBITDA → Cash Tax → CFADS → Senior Debt Service → DSRA → Augmentation Reserve → Cash Available for Distribution → Equity Distribution.
- Dual-asset (data centre + renewable) integration: renewable generation nets against BESS losses and is compared to data centre demand every single period to derive self-sufficiency, shortfall, and grid backup cost.
- Live scenario switcher: a single Cover-sheet cell drives every Assumptions Active-column formula, and every calculation sheet reads only that Active column.
- Debt sculpting: DSCR is held at exactly the 1.35x target in every operating period, and the debt balance fully amortises to zero by the final period, non-circularly.
- Sources = Uses: the Construction tab's total funding sources exactly match total funding uses at COD, to the cent.

8. Glossary

A condensed version of the Appendix-Glossary tab in the workbook, for quick reference alongside this guide.

8.A Technical / Modelling Terms

Term	Definition
CFADS	Cash Flow Available for Debt Service — EBITDA less cash tax; the cash generated before financing costs.
DSCR	Debt Service Coverage Ratio — $\text{CFADS} \div \text{total debt service in a period}$.
LLCR / PLCR	Loan Life / Project Life Coverage Ratio — $\text{PV of remaining CFADS to loan maturity / project life} \div \text{outstanding debt}$.
IDC	Interest During Construction — interest on drawn debt before COD, capitalised into project cost.
DSRA	Debt Service Reserve Account — a cash buffer sized to cover a set number of months of forward debt service.
Debt Sculpting	Structuring principal repayments so debt service tracks a constant multiple of CFADS (the target DSCR) each period, rather than a flat annuity.
PUE	Power Usage Effectiveness — $\text{total facility power} \div \text{IT equipment power}$.
Utilisation	The share of available IT capacity actually in use by customers in a period.
Specific Yield	Annual solar energy generated per unit of installed capacity (kWh/kWp/year).
BESS Round-Trip Efficiency	% of energy retrieved from a battery relative to energy used to charge it.
Circularity	A formula that depends, directly or indirectly, on its own output — e.g. interest → tax → cash flow → debt size → interest.
Gearing	The proportion of total project cost funded by debt rather than equity.

8.B Non-Technical / Commercial Terms

Term	Definition
Take-or-Pay	A contract where the customer pays for contracted capacity whether or not they use it, providing revenue certainty.
Capacity Charge	A fixed monthly fee per kW of contracted data centre capacity, charged regardless of actual usage.
Hyperscale	Very large-scale cloud/data centre operators whose facilities are designed to scale rapidly to very large capacities.
Behind-the-Meter	On-site generation supplying a facility directly, without passing through the public grid meter.

COD	Commercial Operation Date — when the project is complete and begins commercial operations.
SPV	Special Purpose Vehicle — the standalone legal entity that owns, finances, and operates the project.
Limited / Non-Recourse	A financing structure where lenders can only claim against the project's own assets and cash flows, not the sponsors' wider balance sheet.
Senior Debt	Debt ranking first in priority of repayment ahead of other capital.
Equity IRR	The discount rate at which the present value of all equity cash flows nets to zero — the annualised return to equity.
Equity Multiple	Total cash distributed to equity ÷ total equity invested.
Sources & Uses	A reconciliation of how much funding a project needs (Uses) against where it comes from (Sources: debt + equity).
Upfront Fee	A one-off fee paid to lenders (as a % of the facility) at financial close, in addition to the ongoing interest rate.

— End of Guide —